Recovering incidence from a complete neighbourhood

Human Compression Project

Abstract

A transitive group action identifies objects with stabilizer cosets. The missing condition for reconstructing incidence is a converse: every incident pair must be in the diagonal orbit of a chosen base pair. We prove that this is equivalent to one complete neighbourhood calculation for each type pair, and give its finite-cardinality form.

1 The complete reconstruction theorem

Let G act on the right on nonempty sets X_i, transitively on each type. For distinct types, let $R_{ij} \subseteq X_i \times X_j$ be a G-invariant binary relation. Choose objects $x_i \in X_i$, put $H_i = G_{x_i}$, and suppose $x_i R_{ij} x_j$ for every pair of distinct types.

Theorem C.Incidence (equivariant reconstruction). Suppose that, for every $i \neq j$,

$$\{y \in X_j : x_i R_{ij} y\} = x_j^{H_i}. \tag{2.1}$$

Then the maps

$$H_i g \longmapsto x_i^g$$

give an isomorphism from the right-coset incidence structure of the subgroups H_i to the given typed incidence structure. If the neighbour sets are finite, it is enough to check

$$\#\{y \in X_j : x_i R_{ij} y\} = [H_i : H_i \cap H_j]. \tag{2.2}$$

Proof. Equality of right cosets is equivalent to equality of the associated images of x_i, because H_i is exactly its stabilizer. Transitivity gives surjectivity. Two cosets $H_i g, H_j h$ intersect if and only if there is $a \in G$ with $x_i^a = x_i^g$ and $x_j^a = x_j^h$. Thus coset incidence is exactly the diagonal orbit of (x_i, x_j).

The base pair lies in R_{ij}, so invariance puts that entire diagonal orbit inside R_{ij}. Conversely, suppose $x_i^g R_{ij} y$. Applying g^{-1} gives $x_i R_{ij} y^{g^{-1}}$. By (2.1), $y^{g^{-1}} = x_j^u$ for some $u \in H_i$. Then $(x_i^g, y) = (x_i^{ug}, x_j^{ug})$, so the pair lies in the diagonal orbit. This proves equality of the binary relations. All flags and residues are defined by these relations, and therefore are preserved as well.

For the finite criterion, invariance of R_{ij} and the base incidence give $x_j^{H_i} \subseteq R_{ij}(x_i)$. Orbit–stabilizer gives $|x_j^{H_i}| = [H_i : H_i \cap H_j]$. Equality of the two finite cardinalities proves equality of the sets. □

The hypothesis in (2.2) concerns the *complete* neighbour set. Counting a chosen orbit of neighbours would merely repeat orbit–stabilizer and would prove nothing about possible additional incidences. The omitted relation in [HHL12] gives a concrete example of this distinction.

The three Lean declarations `reconstruction`, `localOrbit_subset` and `reconstruction_of_card` in `formal/EquivariantIncidence.lean` formalize the diagonal-orbit argument and the finite-cardinality step, with left actions. Passing to inverse coordinates gives the right-action convention above. They do not formalize the named graph, the whole coset construction, the J1 certificates, or either complete source theorem.

2 Use with the current common theory

For a marking of a finite group, the complete subgroup intersection and typed coset Helly tests provide the hypotheses of Topic C's UC.Geometry. The theorem above then transfers every flag, every residue and every type-preserving automorphism to a source structure whose objects were defined independently. In particular a count of the whole neighbour set is essential; the order of one chosen stabilizer orbit cannot exclude additional incidences. The Livingstone application contains a literal example of exactly this failure.

This is the standard orbit–stabilizer argument, reused from the complete reviewed phase179 proof and its independent phase184 review. It is not a historical-novelty claim. The three unchanged Lean declarations formalize the diagonal-orbit converse and finite-set equality step, not the full coset quotient or the finite group computations. The application retains the complete current Topic C umbrella explicitly; no archived manuscript is a hidden premise.

References

[1] M. I. Hartley, I. Hubard and D. Leemans, *Regular 4-polytopes from the Livingstone graph of Janko's first group*, Journal of Algebraic Combinatorics 35 (2012), 193–214, DOI 10.1007/s10801-011-0300-x.